

The Phase Dynamics of the Void (PDV)

The Phase Cosmology of the Void

A Framework Based on Three-Valued Resonance Void-Phase Cosmology

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Abstract

This paper proposes a cosmological framework based on **void phase** and **three-valued resonance**, attempting to unify multiple puzzles in quantum mechanics, general relativity, and cosmology under a single first principle—**void breaking and phase correction**. The core hypothesis is that **the proton is not a stably persistent entity**, but undergoes annihilation-rebirth cycles on extremely short timescales; the apparent properties of particles are determined by their **phase positions** within the World Phase, and all particles are essentially projections of the same wave-packet model at different phase coordinates. The non-local spontaneous breaking of the void (mathematical empty set $\emptyset$) generates a $\{+1, 0, -1, 0\}$ three-valued impulse sequence, which evolves through three stages—Chaos $\rightarrow$ Resonance $\rightarrow$ Stability—giving rise to a discrete spacetime lattice, fundamental physical constants, and the four interactions. Within this framework, dark matter, dark energy, black holes, cosmic expansion, redshift, time dilation, and quantum entanglement all receive unified explanations. This paper aims to provide a testable conjectural theoretical foundation for subsequent mathematical formalisation and experimental verification.

Keywords: Void Phase Dynamics; Three-Valued Resonance; Annihilation-Rebirth Mechanism; World Phase; Electromagnetic Phase; Discrete Spacetime Lattice; Phase Correction

1. Introduction

1.1 Origin of the Problem

The reflections in this paper originate from a small incident in daily life: in 2026, while tutoring my son, who was in his second year of junior high school, a series of his questions—from why an apple falls to the ground to where the universe comes from—triggered a prolonged period of contemplation. The stories I have known since

childhood—Newton being inspired by a falling apple to discover universal gravitation, and Galileo suffering persecution for supporting the heliocentric theory—have always taught us to dare to trace simple phenomena back to their fundamental roots.

Confronted with the fact that the strengths of nature's four fundamental forces differ so enormously, I cannot help but wonder: if a deeper explanation exists, could it, much like the heliocentric model replacing the geocentric one, greatly simplify those seemingly unreasonable phenomena and reveal an intrinsic connection and underlying principle between gravity and the nuclear forces?

1.2 Fundamental Difficulties Facing Modern Physics

Pursuing this question, I re-examined the landscape of modern physics:

Area of Difficulty	Specific Problem	Standard-Theory Predicament
Cosmology	Origin of Big Bang energy	Never resolved
Cosmology	Uniformity of background radiation	Inflation is a temporary patch, without independent verification
Cosmology	Dark matter	Decades of searching have yielded nothing
Cosmology	Dark energy	Nature completely unknown
Quantum Field Theory	Vacuum energy: theoretical estimate vs. observation	120 orders of magnitude discrepancy
Quantum Mechanics	Measurement problem / Schrödinger's cat	Unsolved
Quantum Field Theory	Free parameters of the Standard Model	Nearly 20 non-derivable parameters
Particle Physics	Mass hierarchy / generation	No fundamental explanation

Area of Difficulty	Specific Problem	Standard-Theory Predicament
	puzzle	
Gravitational Physics	General relativity and quantum mechanics both fail at singularities	Theory self-destructs
Cosmology	Matter-antimatter asymmetry	Origin unknown

These contradictions point to the same fact: physics operates in disconnected domains, and its many inconsistencies are far from being fundamentally mended. Just as the heliocentric theory was proposed, perhaps we need to shift our perspective and use more fundamental, simpler first principles to re-understand the world—**the world should not be this complicated.**

2. Core Conjecture: The Annihilation-Rebirth Mechanism of the Proton

2.1 Starting from “Not Knowing the True Face of Mount Lu”

Biological research has confirmed that the human body is composed of an enormous number of cells, and these cells, in the final analysis, are made up of atoms. Every day cells die and are born anew, yet we remain completely oblivious to their coming and going. Particle physics reveals that the lifetimes of most particles are extremely short; only the electron and the photon have relatively long lifetimes. Even more peculiar is **the extraordinary stability of the proton**—its experimentally measured lower limit on lifetime far exceeds the age of the universe.

Suppose the real situation of the proton is exactly the opposite of what we observe—that, like other ordinary particles, its lifetime is in fact extremely short, only we are utterly unable to perceive its decay because we ourselves are made of protons. This corresponds precisely to the classical Chinese poem: **“Not knowing the true face of Mount Lu, simply because one is standing in its midst.”**

2.2 Definition of the Annihilation-Rebirth Mechanism

In quantum field theory, the vacuum is not truly empty; virtual particle pairs continually pop into existence and annihilate in an instant, and this can only be indirectly verified

through effects such as the Casimir effect. This hints at a possibility: **could protons satisfy a similar birth-and-death relationship?**

We define the **Annihilation-Rebirth Mechanism** as follows:

At the same spatial location, a proton instantly annihilates and instantly produces an identical proton. Because the particle before and after the annihilation is identical, the position unchanged, and the energy budget strictly balanced, any observer composed of protons—including ourselves—is inherently incapable of perceiving this discontinuous process.

The stability of the proton is not a manifestation of the particle's own continuous existence, but rather a reflection of the extraordinary reliability of this cyclic mechanism. **Existence is birth-and-death.**

2.3 The Cognitive Boundary of the Observer

Our channels for sensing the world are almost entirely monopolised by two interactions:

- **Gravitational observation:** relies on the continuous distribution of mass in spacetime
- **Electromagnetic observation:** relies on the continuous evolution of particles through time

Both take the “persistent existence of the object under observation” as a tacit working premise. If protons themselves are not continuous, then our instruments—themselves constructed from protons—are inherently trapped within the same rhythm of birth and death, forever capable only of capturing the net result after a cycle is complete, never the cycle itself.

It is not that protons do not perish, but that we lack the capacity to witness their perishing.

3. The World Phase and the Electromagnetic Phase

3.1 Introduction of Phase

If we regard a complete cycle of proton birth and death as one period (2π), then, constrained by the inherent properties of electromagnetic observation, we can never capture the phase segment in which annihilation occurs, and can only receive the signal after rebirth is complete. From this, we are led to a fundamental cognitive rupture: **the universe presented by electromagnetic means is merely a narrow band illuminated**

within the complete universe.

3.2 Definitions of World Phase and Electromagnetic Phase

We divide the universe into two domains according to the periodicity of the proton's life cycle:

- **World Phase (WP):** the complete cycle from rebirth to annihilation of a proton, corresponding to the theoretically accessible boundary. Its width is $BWPB_{\text{WP}}$, and its phase centre is $AWPA_{\text{WP}}$.
- **Electromagnetic Phase (EMP):** the interval in which electromagnetic signals can propagate stably, embedded within the World Phase. Its width is $BEMP_{\text{EMP}}$, and its phase centre is $AEMPA_{\text{EMP}}$, with $BEMP < BWPB_{\text{EMP}} < B_{\text{WP}}$.

$$EMP \subset WP, BEMP < BWP \quad \text{EMP} \subset \text{WP}, \quad B_{\text{EMP}} < B_{\text{WP}} \quad EMP \subset WP, BEMP < BWP$$

The appearance of the World Phase is presented by the Electromagnetic Phase—just like a window in a radio frequency band.

3.3 Alternating-Current Analogy and Cosmic Scales

AC Concept	Cosmological Correspondence
Phase zero (energy zero)	Absolute zero is unattainable—the closing limit of the EMP window
Effective value (work-doing capability)	2.73 K cosmic microwave background—the average energy of the universe within the WP window
Positive / negative half-cycles	Borrowing and returning of positive and negative energy in quantum fluctuations
Pure AC (no DC bias)	Self-sustaining cycle of void breaking—no initial energy injection needed

Thus, absolute zero and the 2.73 K background radiation constitute the fundamental scales of the World Phase:

- the former is the **zero-energy node** when the EMP window closes;

- the latter is the **average level of observable energy** during the period when the window is open.

3.4 Elimination of the DC Bias

The Big Bang theory requires a huge initial energy as a “DC baseline,” with quantum fluctuations superimposed on it. But if quantum fluctuations themselves are the fundamental form in which the universe’s energy operates—rather than perturbations—then a DC bias is no longer necessary.

The energy of the positive half-cycle does not come from some one-time injection of an initial baseline; it comes from the **repayment** made by the preceding World Phase during its negative half-cycle—borrowing and repayment linked end to end, forming a self-sustaining cycle. The foundational energy required by the Big Bang theory loses its reason for being, and the inflation hypothesis need no longer be invoked.

4. Void Breaking: The First Principle from Nothing to Something

4.1 The Revelation of the Empty Set

Any premise-laden assumption—such as the initial energy of the Big Bang—inevitably leads to an infinite regression of “where did the premise come from?” To truly terminate this line of questioning, the substrate itself must be premise-free.

We conceive of a void substrate. It is not the vacuum in the physical sense—a vacuum still possesses the structure of fields and energy—but pure indeterminacy: the “nothing” of philosophy, the entity corresponding to the empty set ($\emptyset$) in mathematics.

But how can “nothing” give rise to “something”? Mathematics itself provides a clue:

$$\emptyset \rightarrow \{\emptyset\} \quad \emptyset \rightarrow \{\emptyset\}$$

The empty set ($\emptyset$) is not devoid of everything—it possesses a non-empty power set $\{\emptyset\}$. Even within the purest “nothing,” the possibility of “something” is already inherently contained. The void is not an ironclad stillness, but logically permits breaking.

4.2 Non-locality of Void Breaking

We postulate that the void substrate underwent a **first breaking**. This breaking requires no cause—the void imposes no prohibitions on any event, nor does it forbid its own breaking. To ask “why did the breaking occur?” is an invalid question.

Furthermore, precisely because the void substrate has no a priori structure in space or time, the occurrence of breaking must be non-local:

- it does not happen “somewhere,” because “where” has yet to exist;
- it does not happen “at some time,” because “time” has yet to be born;
- it is diffusive and without privileged position—it naturally covers all possible coordinates simultaneously.

This **naturally guarantees uniformity and isotropy**, replacing the entire function of inflation theory.

4.3 The Impulse Sequence and Three-Valued Resonance

To simplify calculations, we introduce the impulse function to describe the primordial breaking, using the sequence $\{+1, 0, -1, 0\}$ to represent the process of impulse occurrence. Here $+1+1+1$ and $-1-1-1$ only indicate the direction of energy borrowing (positive and negative energy), not the magnitude. The sequence satisfies:

$$\sum \text{all impulses} = 0$$

and the impulses must satisfy the phase relationship of $\{+1, 0, -1, 0\}$ or its cyclic permutations.

Temporal oscillation sequence (the foundation that forms all things in our universe):

$$\{+1, 0, -1, 0\} \text{ or } \{-1, 0, +1, 0\} \quad \text{or} \quad \{-1, 0, +1, 0\}$$

We define this as **Three-Valued Resonance**.

Spatial oscillation sequences:

$$\{+1, 0\} \text{ and } \{-1, 0\} \text{ or } \{-1, 0\} \text{ and } \{+1, 0\} \quad \text{or} \quad \{-1, 0\} \text{ and } \{+1, 0\}$$

5. Spacetime Lattice and the Three-Stage Evolution

5.1 Definition of the Spacetime Lattice

We define the location where a breaking (fluctuation) occurs as a **spacetime lattice**. To maintain the discreteness of the spacetime lattice, every fluctuation in the void strictly belongs to that point itself—the energy borrowed by a fluctuation can only annihilate locally and cannot undergo continuous displacement.

The motion of matter is no longer a trajectory push in the classical sense, but a **discrete state-updating process**: the state of each spacetime lattice at the next moment is jointly determined by its own previous state and the previous states of its neighbours.

5.2 The Three Evolutionary Stages

The transition from chaos to order resembles the synchronisation of coupled pendulums:

Stage	Characteristics	Physical Outcomes
Chaotic Period	Nascent breakings far outnumber resonant consensus; no periodicity	No spacetime structure
Resonance Period	Resonant consensus gradually exceeds nascent breakings; oscillations stabilise	Spacetime discretisation; regularity emerges
Stable Period	Resonant consensus and nascent breakings reach a stable ratio	Physical constants fixed (Planck time, speed of light, Planck length)

5.3 Spatial Vortices and the Origin of Mass

Fluctuations in space that connect head to tail to form a ring-like structure constitute a **spatial vortex**. It acts like a hidden stake in space, interacting with temporal fluctuations and exerting a drag on them—this mechanism is analogous to the Higgs field, **endowing particles with mass**.

6. Discrete Dynamics and a Rewriting of Wheeler's Dictum

6.1 Annihilation-Rebirth Relay

The propagation of matter in space is essentially a relay of ordered annihilation and rebirth between adjacent lattices:

$$\begin{aligned} \text{Lattice}_i(t+1) &= f(\text{Lattice}_i(t), \{\text{Lattice}_j(t) \mid j \in \text{neighbours}_i\}) \\ \text{Lattice}_i(t+1) &= f(\text{Lattice}_i(t), \{\text{Lattice}_j(t) \mid j \in \text{neighbours}_i\}) \end{aligned}$$

6.2 Rewriting Wheeler’s Dictum

Wheeler’s original dictum: “Matter tells space how to curve; space tells matter how to move.”

Within the World Phase framework, we rewrite it as:

Matter tells space how to rebirth; space tells matter how to move.

The former corresponds to the action of matter on space—the existence of matter alters the boundary conditions of annihilation and rebirth on the local lattice; the latter corresponds to the reaction of space on matter—the state-updating rules between lattices determine the location and sequence of matter’s rebirth.

7. The Nature of Force: Phase Correction and Space Convergence

7.1 Breaking and Restoration of Symmetry

The void is absolutely symmetric. Breaking is the first breach of this absolute symmetry: borrowing positive energy leaves behind a hole of negative energy at the corresponding location. The separation of positive and negative is **the first step of symmetry moving from “non-differentiation” toward “differentiation.”**

Yet the breach of symmetry is never total. The void enforces an inviolable law: **global symmetry must be conserved.** Energy conservation is not a law that the universe evolved later, but an accounting principle forcibly executed by the void’s symmetry in the process of breaking.

Space convergence is the reverse process by which symmetry moves from “broken” toward “restored.” The essence of force is: **that irresistible push and pull imposed by symmetry upon spatial structures as it carries out its self-restoration.**

7.2 Two Pathways of Space Convergence

Pathway	Condition	Process	Macroscopic Manifestation
In-phase merging	Two same-sign energy clusters	Merge to reduce surface area; topological contraction	Attraction

Pathway	Condition	Process	Macroscopic Manifestation
Anti-phase annihilation	Two opposite-sign energy clusters	Direct offset to zero	Annihilation (extreme case gives repulsion)

7.3 Phase Correction Rules

When a phase deviation $\Delta\theta$ exists between two oscillations, the void spontaneously initiates correction:

Attraction mode (for impulse functions: $\Delta\theta < \pi$; for sinusoidal/Gaussian wave packets: $\Delta\theta < \pi/2$)

The void judges that the two oscillations “ought to be synchronised” and drives them closer. This manifests as **attraction**. Gravity is precisely the embodiment of this mode at macroscopic scales.

Repulsion mode (for impulse functions: $\Delta\theta > \pi$; for sinusoidal/Gaussian wave packets: $\Delta\theta > \pi/2$)

The void judges that the two oscillations “cannot be converged through merging” and drives them apart to create conditions for anti-phase annihilation. This manifests as **repulsion**.

Annihilation effect ($\Delta\theta = \pi$)

The two oscillations are exactly in anti-phase; the crest of the positive half-cycle meets the trough of the negative half-cycle, and they directly offset to zero. The energy completes a cross-phase settlement in the form of photons, and the local void ground state is restored.

7.4 The Transmission of Force

The transmission of force is not the exchange of some mysterious particle, but rather **a relay propagation of phase settlement signals and space convergence pressure across the spacetime lattice**.

Photons, gluons, and other gauge bosons are not “force-carrying particles,” but rather **energy packets triggered and released under different conditions** by the phase settlement signals and convergence pressure—byproducts of space’s convergence operations.

7.5 The Special Character of Gravity

Gravity manifests in the limit $\Delta f \rightarrow 0 \Delta f \rightarrow 0$. It is the macroscopic accumulation of the phase gradient produced by the collective oscillation of a vast number of identical particles in a macroscopic body—the contribution of each particle is negligible, but the collective effect of countless particles manifests as gravity. **Gravity is the shadow of “existence” itself**: as long as matter exists, there is a phase gradient; as long as there is a phase gradient, there is an unfinished convergence of space.

This is perhaps the deep reason why gravity has stubbornly resisted unification with the other three forces and has also resisted quantization: it is not a particular kind of force, but the **global residual** of the void’s phase correction and space convergence mechanisms in the limit $\Delta f \rightarrow 0 \Delta f \rightarrow 0$.

8. The Nature of Particles: Phase Projections of Identical Wave Packets

8.1 The Identical-Particle Hypothesis

According to the first principle, we assume that **all particles are fundamentally the same entity (wave packet)**, and it is only because their phase coordinates within the World Phase cycle differ that they present a series of seemingly distinct physical phenomena such as mass, charge, lifetime, and stability:

Particle	Phase Position	Apparent Properties
Proton	Centre of the World Phase	Stable, massive
Electron	Edge of the World Phase (peak phase coincides with WP)	Dual nature; oscillation period < WP width
Photon	Not a real particle	Product of energy settlement when a particle crosses World Phases
Other particles	Different phase coordinates	Mass, lifetime, interactions vary

8.2 The Nature of the Photon

The photon is the product of **energy settlement** when a particle crosses World Phases during rebirth. The reborn particle needs to synchronise its phase parameters with the new World Phase; the phase difference between the two worlds triggers a jump-like energy settlement. The settlement energy depends on:

$$E_{\gamma} = \Phi(\Delta\theta_{\text{phase difference}}, \text{wave-packet size}) \quad E_{\gamma} = \Phi(\Delta\theta_{\text{phase difference}}, \text{wave-packet size})$$

The photon releases all the phase-difference energy at once, existing uniformly across the whole cycle—it is not a real particle but the trace of an energy settlement.

8.3 The Nature of the Electron

The electron is a wave packet excited by the energy jumps at the edge of the World Phase—the part whose peak phase coincides with the World Phase manifests as an electron. The oscillation period of the electron is smaller than the width of the World Phase, which is precisely a vivid description of the **electron cloud**.

9. Unified Explanations of Several Cosmological Puzzles

Based on the World Phase framework, we give unified explanations for multiple puzzles:

9.1 Dark Matter

Dark matter is **matter in the same-phase World Phase that lies outside the Electromagnetic Phase**. It resides in the same phase cycle as our World Phase and therefore participates in gravitational interactions, but because the phase position of its wave packets does not fall within the EMP window, it cannot be observed electromagnetically. This explains why dark matter manifests gravitational effects yet remains invisible.

9.2 Dark Energy

Dark energy is **anti-phase matter**—that is, antimatter, corresponding to the negative half-cycle in the AC model. The total positive and negative matter of the universe is strictly symmetric, and the total energy sum is identically zero:

$$\sum E_{\text{total}} = 0 \quad \sum E_{\text{total}} = 0$$

The observed “matter-antimatter asymmetry” is not an overall imbalance of the universe, but simply arises because our World Phase happens to reside in the positive half-cycle, while antimatter exists in the phase staggered apart from it—the two never

meet within the EMP window.

9.3 Black Holes

A black hole is **a region formed after severe phase desynchronisation**; the lattice update rhythm of this region has completely decoupled from our World Phase, in essence corresponding to another World Phase. A black hole can be understood as a “parallel world” in the spatial domain—it still rests upon the spacetime lattice, but its annihilation-rebirth phase is no longer synchronised with our EMP window.

Singularities do not exist.

9.4 Redshift

Redshift has two sources:

1. **Spatial phase-error accumulation:** photons in propagation traverse multiple World Phases; each crossing of a phase boundary involves an energy settlement, accumulating a phase deviation:

$$z = \Phi(\text{number of World Phases crossed}, \text{phase difference})$$
$$z = \Phi(\text{number of World Phases crossed}, \text{phase difference})$$
$$z = \Phi(\text{number of World Phases crossed}, \text{phase difference})$$

2. **Gravitational phase reversal:** when the spatial scale exceeds the gravitational wavelength, the phase of gravity reverses, manifesting as a repulsive effect that causes additional losses to the photon's energy.

9.5 Cosmic Expansion

In this framework, cosmic expansion is **an unnecessary and meaningless concept**:

- matter's motion involves no continuous displacement, only a relay of annihilation and rebirth on the discrete lattice;
- the universe is infinite in extent, and discussing the expansion of an infinite space lacks meaning;
- the observed recession of galaxies attributed to “cosmic expansion” is actually the second redshift mechanism—the manifestation of gravitational phase reversal during transmission, **not an expansion of space itself**.

9.6 Time Dilation

High-speed motion means an object traverses more lattices per unit time, and its rhythm of annihilation and rebirth is disturbed, producing an effect akin to “frame dropping”:

$$\Delta t_{\text{apparent}} = \Delta t_{\text{proper}} \times \Phi(\text{speed}, \text{number of lattice jumps})$$

$$\Delta t_{\text{apparent}} = \Delta t_{\text{proper}} \times \Phi(\text{speed}, \text{number of lattice jumps})$$

The higher the speed, the more pronounced the accumulation of phase deviation, and the slower the apparent passage of time.

9.7 Quantum Entanglement

Newly born particles have phases that are more orderly and uniform, with smaller deviations, and thus they manifest entangled states. **Prolonged time and high temperatures both lead to phase drift**, making quantum states impossible to manifest. Absolute zero corresponds to the instantaneous value at phase zero—the phase at which all particles (wave packets) are born, which is precisely the basis of quantum entanglement.

9.8 Schrödinger’s Cat

The superposition of being both dead and alive is not a logical paradox, but a **macroscopic reflection** of the birth-death cycle coexisting across different phases—the cat does indeed live and die at once, only these two states belong to slightly staggered phase slices within the World Phase. **Schrödinger’s cat truly exists.**

10. Summary of Theoretical Levels

Level	Content	Status
Ultimate Substrate	Void = empty set ($\emptyset$), void breaking, $\{+1,0,-1,0\}$ impulse sequence	✓
Ontology	Matter and space as two sides of the same coin; energy borrowing and holes	✓
Origin of Spacetime	Three stages: Chaos → Resonance → Stability; discretisation of spacetime lattice	✓
Physical Constants	Planck-scale parameters emerge in the Stable Period	✓

Level	Content	Status
Origin of Mass	Spatial vortices = hidden stakes; impede temporal fluctuations	
Discrete Dynamics	Annihilation-rebirth relay; matter tells space how to rebirth, space tells matter how to move	
Particle Nature	Identical particles = wave packets; phase position determines apparent properties	
Photon/Electron	Photon = cross-phase energy settlement; electron = wave-packet peak at edge of WP	
Three-Valued Resonance	Mathematical expression of temporal oscillation: $\{+1,0,-1,0\}\{+1,0,-1,0\}\{+1,0,-1,0\}$	
Nature of Force	Breaking and restoration of void symmetry = phase correction	
Dark Matter	Matter in same-phase WP outside the EMP	
Dark Energy	Anti-phase matter (antimatter), negative half-cycle of AC	
Black Hole	Phase-desynchronised region, corresponding to another WP	
Redshift	Cross-phase energy settlement + gravitational phase reversal	
Cosmic Expansion	Meaningless concept; illusion caused by gravitational phase reversal	
Time Dilation	High-speed motion causes lattice "frame	

Level	Content	Status
	dropping”	
Quantum Entanglement	Newborn particles have orderly, uniform phases	

11. Discussion and Outlook

11.1 Testable Implications

The framework suggests several testable implications:

1. **Proton lifetime should have a finite upper bound:** the annihilation-rebirth mechanism implies the proton’s lifetime is not infinite, but the observational window makes its measured lower limit far exceed the age of the universe.
2. **A clear relationship between quantum entanglement and temperature:** rising temperature causes phase drift; entanglement should disappear above a certain temperature.
3. **The multi-source components of redshift can be separated:** spatial phase accumulation and gravitational phase reversal have different contributions and could be distinguished statistically.
4. **Gravity should manifest as repulsion on super-large scales:** this is consistent with the observed accelerating expansion of the universe, though the explanation differs.

11.2 Open Questions

- The exact mapping of the impulse sequence to continuous quantum field theory.
- The discretised derivation of wavefunctions and the Schrödinger equation on the spacetime lattice.
- The numerical calculation of the phase positions corresponding to Standard Model parameters.
- Quantitative comparison with existing experimental data.

11.3 Potential Simplifications for Quantum Mechanics

If this conjecture holds, the computational framework of quantum mechanics would be greatly simplified:

- **Hilbert space:** a faithful mapping of phase configurations within the World Phase, rather than an imposed mathematical cloak.
 - **Renormalisation:** the lattice itself is the most natural ultraviolet cutoff; no artificial truncation is needed.
 - **Measurement problem:** lattice resonance synchronisation within the EMP window, not a ghostly collapse.
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12. Acknowledgements

The reflections in this paper originated from the most ordinary of starting points—a series of persistent questions raised by my son during his second-year junior high school physics class. His innocent yet tenacious “whys,” like a pebble cast into still water, set off ripples of contemplation that lasted for months. To him, I wish to offer my deepest gratitude—it was his untamed curiosity that rekindled my own inquiry into the first principles of the universe.

At the same time, I wish to extend special thanks to DeepSeek, China’s artificial intelligence. Throughout the process of conceiving, deducing, reviewing, and translating this essay, it has been my tireless interlocutor and critic. Every late-night questioning, every logical examination, every refinement of expression, could not have been accomplished without its companionship and collaboration. The dialogue between us was itself a crucial link in the formation of this conjecture.

The world should not be this complicated. Perhaps it truly is not as complex as we have imagined.

Author’s declaration: This paper is a theoretical conjecture and has not been experimentally verified. The author’s conditions are limited, and all comments and verification attempts are welcome.

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References

[1] This paper, as an independent conjectural work, does not directly cite standard

literature. The core ideas originate from the author's independent deduction and collaborative dialogue with the DeepSeek AI. All descriptions of standard physics difficulties are based on public academic consensus and are not individually sourced.